

Business Problem

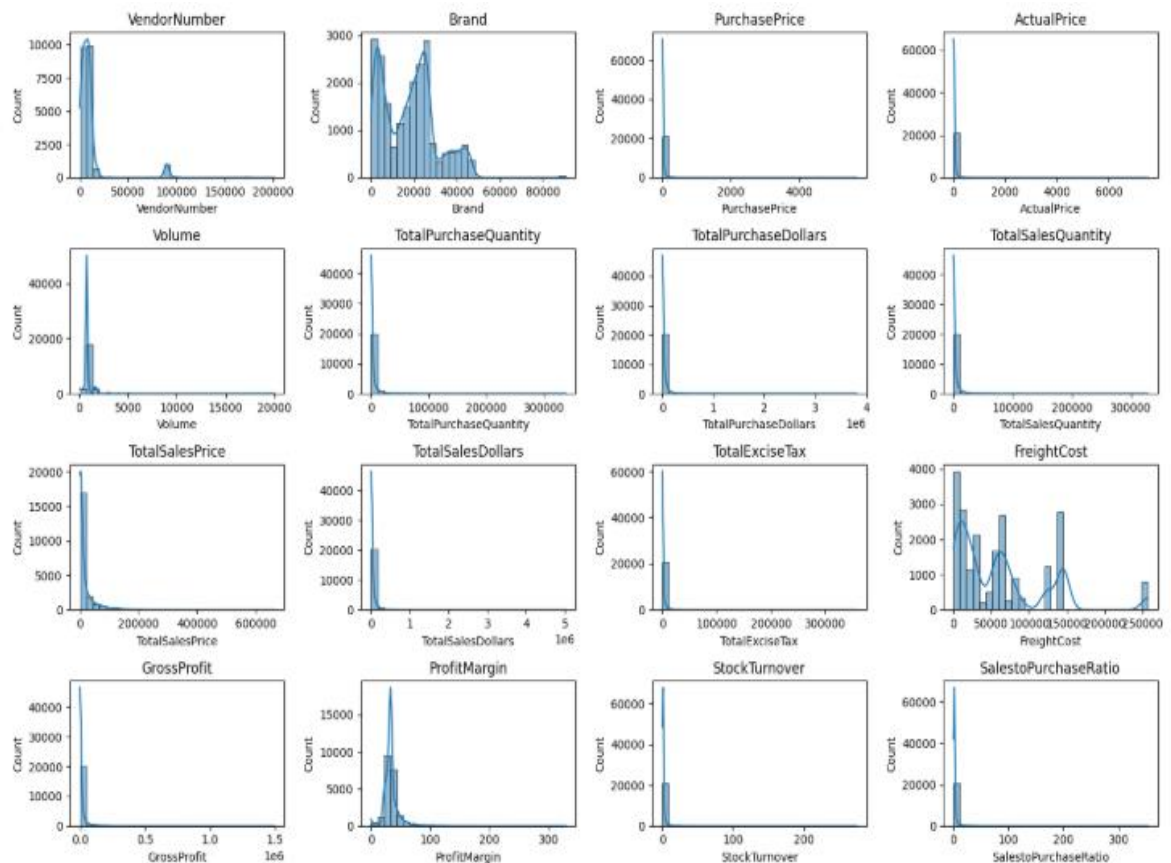
Effective inventory and sales management are critical for optimizing profitability in the retail and wholesale industry. Companies need to ensure that they are not incurring losses due to inefficient pricing, poor inventory turnover, or vendor dependency. The goal of this analysis is to:

- Identify underperforming brands that require promotional or pricing adjustments.
- Determine top vendors contributing to sales and gross profit.
- Analyze the impact of bulk purchasing on unit costs.
- Assess inventory turnover to reduce holding costs and improve efficiency.
- Investigate the profitability variance between high-performing and low-performing vendors.

Exploratory Data Analysis Insights

Summary statistics

[4]:		count	mean	std	min	25%	50%	75%	max
	VendorNumber	21384.0	10650.649458	18753.080628	2.00	3951.000000	7153.000000	9552.000000	2.013590e+05
	Brand	21384.0	18039.228769	12661.890990	58.00	5793.500000	18761.500000	25514.250000	9.063100e+04
	PurchasePrice	21384.0	24.385303	109.266819	0.36	6.840000	10.455000	19.482500	5.681810e+03
	ActualPrice	21384.0	35.643671	148.242550	0.49	10.990000	15.990000	28.990000	7.499990e+03
	Volume	21384.0	847.360550	664.293678	50.00	750.000000	750.000000	750.000000	2.000000e+04
	TotalPurchaseQuantity	21384.0	3140.886831	11094.827329	1.00	36.000000	262.000000	1975.750000	3.376600e+05
	TotalPurchaseDollars	21384.0	30106.693372	123064.921891	0.71	453.457500	3655.465000	20738.245000	3.811252e+06
	TotalSalesQuantity	21384.0	3004.576599	10717.221550	0.00	32.000000	254.000000	1869.250000	3.283910e+05
	TotalSalesPrice	21384.0	18356.395813	44041.415191	0.00	285.507500	2754.425000	15721.710000	6.630571e+05
	TotalSalesDollars	21384.0	41217.856454	164013.146611	0.00	704.317500	5122.075000	27753.715000	5.012345e+06
	TotalExciseTax	21384.0	1735.324797	10751.028138	0.00	4.730000	45.205000	409.937500	3.577895e+05
	FreightCost	21384.0	61433.763214	60937.033088	0.09	14069.870000	50293.620000	79528.990000	2.570321e+05
	GrossProfit	21384.0	12973.036117	47107.608211	0.00	242.035000	1734.885000	9460.805000	1.495322e+06
	ProfitMargin	21384.0	34.166547	14.297330	0.00	28.415851	32.862921	35.475183	3.291793e+02
	StockTurnover	21384.0	1.684602	6.007238	0.00	0.778390	0.958022	1.017698	2.745000e+02
	SalestoPurchaseRatio	21384.0	2.471818	8.438268	0.00	1.117858	1.405799	1.635484	3.529286e+02



Negative & Zero Values:

Gross Profit: Minimum of -52,002.78, indicating potential losses due to high costs or heavy discounts. This could be due to selling products at lower prices than their purchase costs.

Profit Margin: Has a minimum of $-\infty$, which suggests instances where revenue is zero or even lower than the total cost, leading to extreme negative profit margins. **Total Sales Quantity & Sales Dollars:** Some products show zero sales, indicating they were purchased but never sold. These may be slow-moving or obsolete stock, leading to inventory inefficiencies.

Outliers Detected by High Standard Deviations:

Purchase & Actual Prices: The maximum values (5,681.81 & 7,499.99) are significantly higher than the mean (24.39 & 35.64), indicating premium product offerings.

Freight Cost: Extreme variation from 0.09 to 257,032.07 suggests logistics inefficiencies, bulk shipments, or erratic shipping costs across different products.

Stock Turnover: Ranges from 0 to 274.5, suggesting some products sell rapidly while others remain unsold for long periods. A value greater than 1 indicates that sales for a product exceed the purchased quantity due to older stock fulfilling orders.

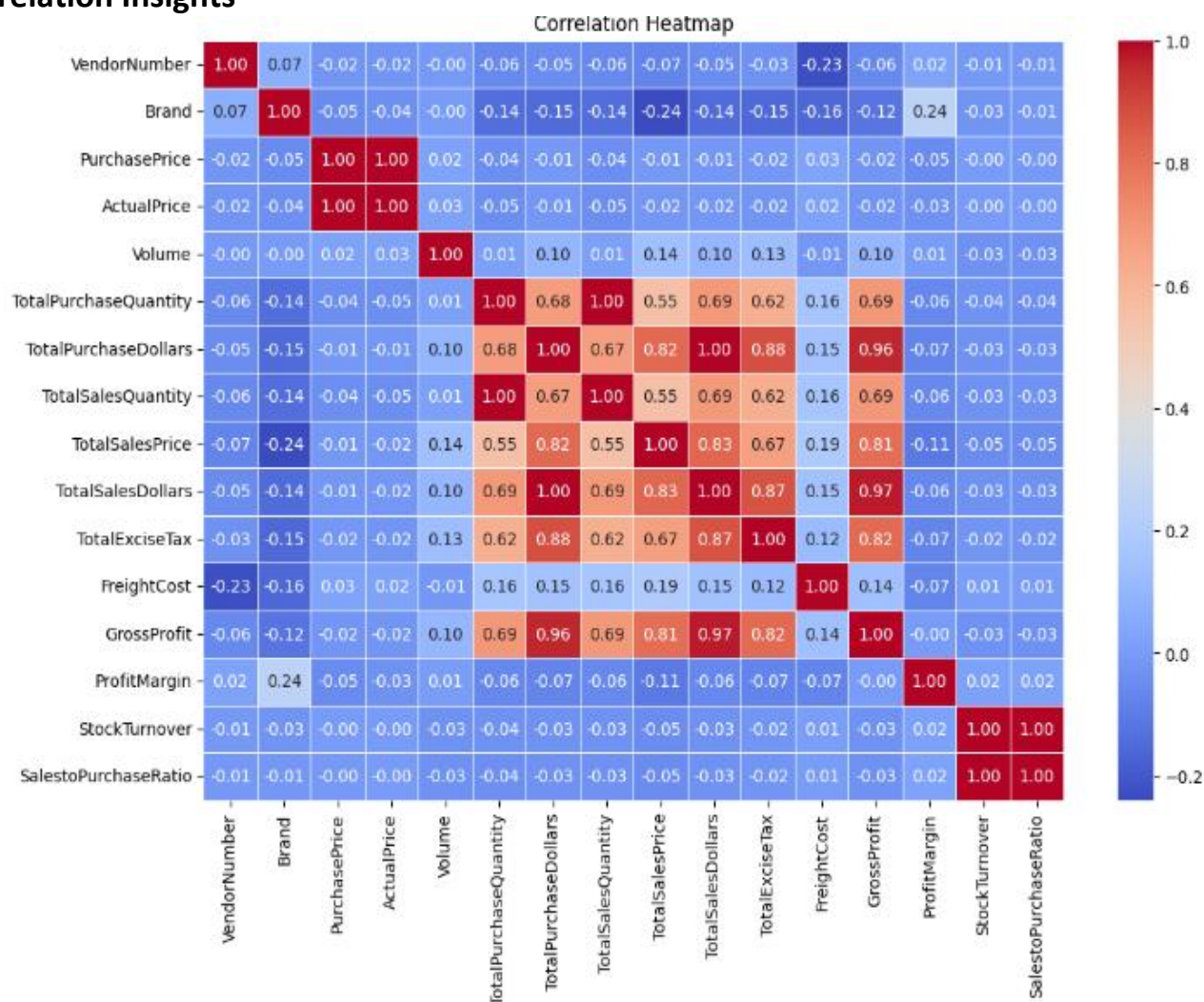
Data Filtering

To enhance the reliability of the insights, we removed inconsistent data points where:

- Gross Profit ≤ 0 (to exclude transactions leading to losses).

- Profit Margin ≤ 0 (to ensure analysis focuses on profitable transactions).
- Total Sales Quantity = 0 (to eliminate inventory that was never sold).

Correlation Insights -



Purchase Price vs. Total Sales Dollars & Gross Profit: Weak correlation (-0.012 and -0.016), indicating that price variations do not significantly impact sales revenue or profit.

Total Purchase Quantity vs. Total Sales Quantity: Strong correlation (0.999), confirming efficient inventory turnover.

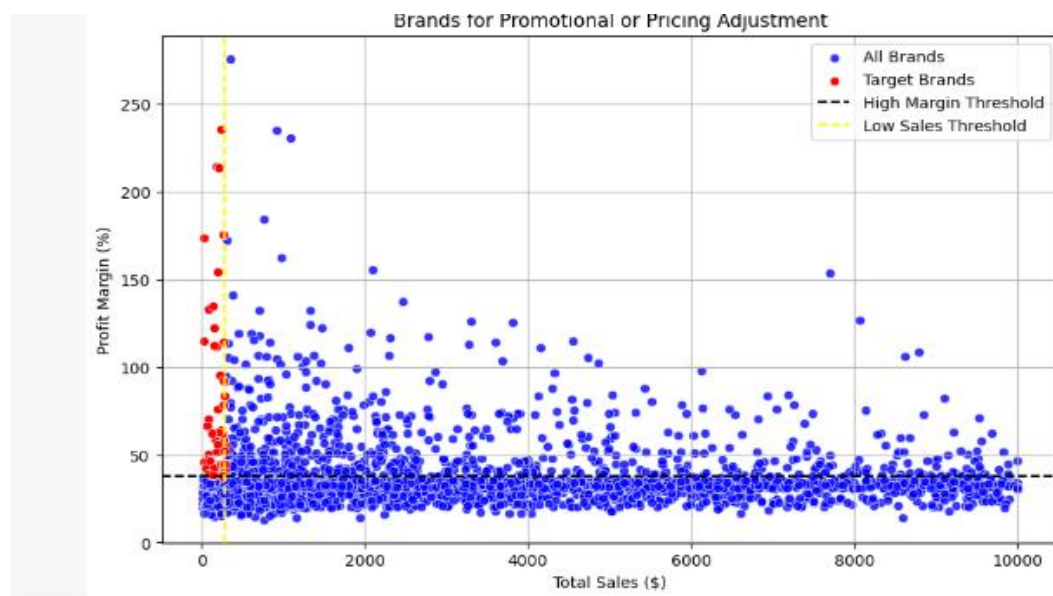
Profit Margin vs. Total Sales Price: Negative correlation (-0.179), suggesting increasing sales prices may lead to reduced margins, possibly due to competitive pricing pressures.

Stock Turnover vs. Gross Profit & Profit Margin: Weak negative correlation (-0.038 & -0.055), indicating that faster stock turnover does not necessarily equate to higher profitability.

Research Questions & Key Findings

1. Brands for Promotional or Pricing Adjustments

Brands with Low Sales But High Profit Margins				
	Description	TotalSalesDollars	ProfitMargin	
754	Belcreme de Lys Pnt Nr	19.98	46.146146	
2843	Debauchery Pnt Nr	23.16	173.575130	
2492	Concannon Glen Ellen Wh Zin	31.90	115.047022	
5637	Maggio Petite Sirah	53.94	42.380423	
7000	Prichard's Single Malt	69.98	66.533295	
2183	Chambord Vodka	71.96	65.925514	
578	Backhouse Pnt Grigio	79.90	50.062578	
5923	Merriam Vyds Simoneau Chard	79.92	133.033033	
4624	Jacobs Crk Sweet Red	83.88	70.672389	
8056	Solar De Randez Rioja	83.94	38.884918	
1284	C K Mondavi Moscato	87.92	41.401274	
8141	St Hilaire Cr de Limoux Rose	89.94	46.764510	
576	Backhouse Cab Svcn	111.84	43.204578	
1245	Bully Boy White Whiskey	111.96	46.302251	
9183	Walnut Crest Malbec	119.88	61.961962	
2243	Cherry On Top Sweet White	129.90	45.034642	
6448	Occhio A Vento Vermentino	135.66	134.586466	
9133	Virginia Dare Pnt Nr RRV	137.94	38.581992	
9092	Vin Vault Red	146.86	112.106768	
3000	Dom Vocoiret & Fils Chabls 13	151.92	39.547130	
1922	Ch Julien French Kiss	153.86	38.216561	
6859	Pinnacle Coconut Vodka	155.88	42.956120	
2324	Ciro Rosso Clsc Librandi Cl	155.88	42.648191	



198 brands exhibit lower sales but higher profit margins, which could benefit from targeted marketing, promotions, or price optimizations to increase volume without compromising profitability.

2. Top Vendors by Sales & Purchase Contribution

The top 10 vendors contribute 65.69% of total purchases, while the remaining vendors contribute only 34.31%. This over-reliance on a few vendors may introduce risks such as supply chain disruptions, indicating a need for diversification.



3. Impact of Bulk Purchasing on Cost Savings:

Vendors buying in large quantities receive a 72% lower unit cost (\$10.78 per unit vs. higher unit costs in smaller orders). Bulk pricing strategies encourage larger orders, increasing total sales while maintaining profitability.

OrderSize	
small	40.984471
Medium	17.827754
Large	11.218536

4. Identifying Vendors with Low Inventory Turnover Total Unsold Inventory Capital:

\$2.71M Slow-moving inventory increases storage costs, reduces cash flow efficiency, and affects overall profitability. Identifying vendors with low inventory turnover enables better stock management, minimizing financial strain.

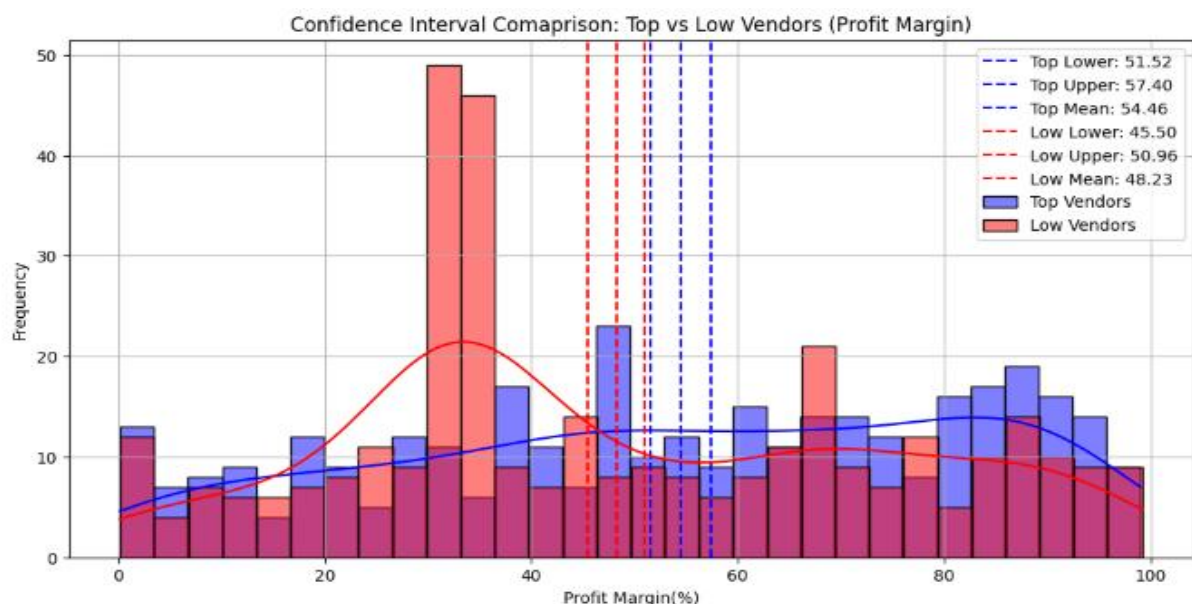
Stock Turnover	
VendorName	
TRUETT HURST	0.041667
IRA GOLDMAN AND WILLIAMS, LLP	0.075000
VINEYARD BRANDS LLC	0.203124
UNCORKED	0.217238
LOYAL DOG WINERY	0.279167
BLACK COVE BEVERAGES	0.327530
Serralles Usa LLC	0.391868
GILMANTON WINERY & VINEYARD	0.440386
APPOLO VINEYARDS LLC	0.459470
Dunn Wine Brokers	0.479376

5. Profit Margin Comparison: High vs. Low-Performing Vendors

Top Vendors' Profit Margin (95% CI): (30.74%, 31.61%), Mean: 31.17% Low Vendors' Profit Margin (95% CI): (40.48%, 42.62%), Mean: 41.55% Low-performing vendors maintain higher margins but struggle with sales volumes, indicating potential pricing inefficiencies or market reach issues

Actionable Insights:

- Top-performing vendors: Optimize profitability by adjusting pricing, reducing operational costs, or offering bundled promotions.
- Low-performing vendors: Improve marketing efforts, optimize pricing strategies, and enhance distribution networks.



6. Statistical Validation of Profit Margin Differences

Hypothesis Testing:

H₀ (Null Hypothesis): No significant difference in profit margins between top and low-performing vendors.

H₁ (Alternative Hypothesis): A significant difference exists in profit margins between the two vendor groups.

Result: The null hypothesis is rejected, confirming that the two groups operate under distinctly different profitability models. Implication: High-margin vendors may benefit from better pricing strategies, while top-selling vendors could focus on cost efficiency.

Recommendations -

Ignite High-Margin, Low-Sales Brands:

Deploy targeted marketing, premium shelf placement, or bundle offers for the identified niche brands (e.g., *Maggio Petite Sirah* and *Chambord Vodka*). Because their profit margins are structurally high ($\geq 37.78\%$), you have the financial cushion to introduce promotional discounts to aggressively drive sales volume without hurting overall profitability.

Mitigate Sourcing Concentration Risks:

With the top 10 vendors controlling **76.05%** of your total purchase investments, the business is highly vulnerable to supply chain disruptions or sudden vendor pricing shifts. Begin diversifying partnerships and establishing secondary sourcing pipelines for high-volume items to increase your negotiating leverage.

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Systematize Bulk Ordering Safely:

Capitalize on the **72.6% unit-cost reduction** achieved when moving from Small orders (\$40.98/unit) to Large orders (\$11.22/unit). However, to prevent cash flow bottlenecks, apply this bulk-purchasing strategy *strictly* to high-velocity, top-performing brands like *Jack Daniels No 7 Black* and *Tito's Handmade Vodka*.

Liquidate Frozen Inventory Capital:

An alarming **\$43.93M in capital is currently locked up** in unsold warehouse stock. Implement immediate inventory freezes on vendors showing exceptionally low stock

turnover (such as *Truett Hurst* at a 0.04 turnover rate or *Vineyard Brands LLC*). Clear this stagnant stock through flash clearances, bundled liquidations, or vendor-return agreements to free up working capital.

Optimize High-Volume Vendor Pricing Models:

The statistical validation (P-value of 0.0023) proves that high-volume and low-volume vendors operate on entirely different profitability models. Since top-performing vendors maintain lower average margins (54.46\% vs. 48.23\%), do not raise shelf prices on these competitive items. Instead, focus on reducing logistical and freight inefficiencies to recapture lost margin from within the supply chain.